

Aakash Yadav

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Professional Summary

Fresh Computer Engineering graduate who builds and ships from **real-time collaborative apps** to **AI-powered recruitment agents**. I don't wait to learn. I **learn by doing and document the journey publicly on Twitter daily**. Comfortable across the full stack and **actively expanding into AI agents and LLM tooling**.

Education

JC BOSE UNIVERSITY OF SCIENCE AND TECHNOLOGY , YMCA – Faridabad 7.98 GPA (2026)
Computer Engineering

Projects

CollabxDraw - RealTime Collaborative Whiteboard Present

- Built and deployed a production-grade collaborative whiteboard where multiple users draw simultaneously in real time, with changes propagated to all connected clients via persistent WebSocket connections.
- Engineered a dual-backend architecture — Express REST API handles JWT auth, bcrypt-secured room passwords, and initial canvas hydration; a dedicated WebSocket server manages live event broadcasting across sessions.
- Optimized perceived load time by pre-fetching full canvas state over HTTP before WebSocket handshake, eliminating blank-canvas flicker on room join.
- Structured codebase as a Turborepo monorepo (3 apps, 3 shared packages) with pnpm workspaces, enabling shared Zod schemas, auth utilities.
- Containerized all services with multi-stage Docker builds for environment-consistent deployment; frontend shipped via Vercel.

Avni - AI Recruitment Agent 02 Feb – 23 May 25

- Built an end-to-end AI recruitment agent that semantically scores resumes against job descriptions, identifies skill gaps, and generates optimized resume rewrites.
- Implemented a RAG pipeline using LangChain + FAISS vector store: resumes are chunked via RecursiveCharacterTextSplitter, embedded with OpenAI Embeddings, and retrieved via semantic similarity for context-aware Q&A and scoring.
- Designed a skill-extraction LLM chain that parses any custom job description to identify target requirements, then runs a RetrievalQA match against candidate resume chunks to produce an automated fit score.
- Generated personalized interview question sets (Coding / System Design / Theoretical) dynamically tailored to each candidate's specific strengths and weaknesses extracted from their resume.
- Delivered paragraph-level resume feedback with before/after rewrite examples and a full resume regeneration feature that fills identified gaps within strict context boundaries.

OwnShell - Custom Unix Shell Interpreter Present

- Building a custom shell interpreter from scratch in TypeScript/Bun, implementing a fully interactive REPL with built-in commands (cd, pwd, echo, type, exit) and external binary execution via \$PATH resolution.
- Wrote a custom command tokenizer and argument parser from the ground up — no shell libraries — handling edge cases in argument splitting and command dispatch.

Skills

Languages: TypeScript, Python, JavaScript

Frontend: React, React Router, NextJS, HTML/CSS

Backend: Node.js, Express.js, WebSockets, REST APIs

AI/ML: LangChain, OpenAI APIs, FAISS, RAG pipelines

Databases: PostgreSQL, Prisma ORM

DevOps/Tooling: Docker, Turborepo, pnpm, Git, Bun, AWS, Linux

Certifications

AI Engineer for Developers Associate Certification – DataCamp